

Compactifications of the configuration spaces of points in $\mathbb{P}^2$

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1 Introduction

The moduli problem of n -distinct points on the projective line $\mathbb{P}^1$ modulo PGL_2 is well studied, with the famous compactification $\overline{\mathcal{M}}_{0,n}$ obtained by Deligne and Mumford. There is a connection between projective point configurations a quotients of the Grassmannian $\mathrm{Gr}(2, n)$, which has two natural compactifications as a Chow quotient and through the theory of matroid polytopes. In the case of points in $\mathbb{P}^1$, these give isomorphic spaces to $\overline{\mathcal{M}}_{0,n}$.

The generalisation to points in $\mathbb{P}^{r-1}$ gives three distinct schemes that are no longer necessarily isomorphic. We first construct the direct generalisation – Gerritzen and Piwek’s compactification $\overline{\mathcal{B}}_n$ for points in $\mathbb{P}^2$, before proving the Gelfand-MacPherson correspondence and discussing the modular interpretation of Lafforgue’s varieties in arbitrary dimension.

Theorem 1. The *Gelfand-MacPherson correspondence* gives an isomorphism between quotients of the configuration space $\mathbf{X}^P(r, n)$ of n hyperplanes in $\mathbb{P}^{r-1}$ corresponding to a matroid polytope $P \subset \Delta(r, n)$, and the corresponding matroid stratum of the Grassmannian

$$\mathbf{X}^P(r, n) \cong \mathrm{Gr}^{P,0}(r, n)/H.$$

where $H = (k^*)^n/k^*$, acting on $\mathrm{Gr}(r, n)$ via its natural action on k^n .

Theorem 2. The Lafforgue variety is a compactification $\overline{\Omega}^P$ of $\mathrm{Gr}^{P,0}(r, n)/H$, stratified by regular subdivisions of the polytope P , which correspond to degenerations of a projective point configuration.

Finally, the morphisms between $\overline{\mathcal{B}}_n, \overline{\Omega}^{\Delta(3,n)}$ and Kapranov’s Chow quotient are discussed, with the following key result:

Theorem 3. There exists a finite birational morphism $\overline{\mathbf{X}}(3, n) \rightarrow \overline{\mathcal{B}}_n$. In particular, $\overline{\mathbf{X}}(3, n)$, $\overline{\mathcal{B}}_n$ and the main stratum of $\overline{\Omega}^{\Delta(3,n)}$ have the same normalisation.

2 $\overline{\mathcal{M}}_{0,n}$ and the Gerritzen-Piwek generalisation

2.1 Motivating the compactification $\overline{\mathcal{M}}_{0,n}$

Fix an algebraically closed field k with characteristic zero, writing $\mathbb{P}^m := \mathbb{P}_k^m$. The problem we are concerned with here is the moduli of n ordered lines through the origin in k^r , where two configurations are considered the same if they are equivalent up to a projective coordinate change (a PGL_r action on $\mathbb{P}^{r-1}$). Naively then, since each line corresponds to a point in $\mathbb{P}^{r-1}$, our moduli space is $(\mathbb{P}^{r-1})^n/\mathrm{PGL}_r$.

The quotient $(\mathbb{P}^{r-1})^n/\mathrm{PGL}_r$ is a perfectly reasonable space, but has a few properties that we don’t desire. The main one is that the PGL_r action on $(\mathbb{P}^{r-1})^n$ is not free: if all n lines lie in some hyperplane

then they are fixed by a non-trivial subgroup of PGL_r , which suggests that the symmetry group may be too large, or that the space we are quotienting is not correct.

The viewpoint taken here is the latter. Consider the moduli space of n *general* ordered points in $\mathbb{P}^{r-1}$ up to a PGL_r action, which is (isomorphic to) a dense open subset of $(\mathbb{P}^{r-1})^n$, modulo PGL_r . Here general means no r -element subset lies in a hyperplane, which is a sufficient condition for the PGL_r action to be free. In particular, for $r = 2$ this is $M_{0,n}$, the moduli space of n distinct points in $\mathbb{P}^1$ up to a PGL_2 action (since the hyperplanes in $\mathbb{P}^1$ are points). This moduli space is not proper (the algebro-geometric notion of compact), however, and so we seek a compactification that captures the geometric information of limiting families of points.

For $M_{0,n}$, the Deligne-Mumford compactification $\overline{M}_{0,n}$ is well studied, and can be realised in many ways. The adjoined points in the boundary $\partial\overline{M}_{0,n} = \overline{M}_{0,n} \setminus M_{0,n}$ correspond to *stable* n -pointed genus zero curves, which are algebraically isomorphic to copies of $\mathbb{P}^1$ identified at points. Their automorphism group is no longer only PGL_2 , but rather a product of copies of PGL_2 that act on each copy of $\mathbb{P}^1$ individually, and an n -point configuration on a curve C is stable when the $\mathrm{Aut}(C)$ action on the point configuration is free (which concretely corresponds to three marked points being on each copy of $\mathbb{P}^1$, since an element of PGL_2 is uniquely determined by where it sends three points). Importantly, these adjoined points do not correspond to degenerate point configurations in $\mathbb{P}^1$, but rather to non-degenerate configurations on a degeneration of the ambient space $\mathbb{P}^1$.

We can concretely construct the compactification $\overline{M}_{0,n}$ as follows: before the PGL_2 quotient, configurations correspond to points in $\tilde{U}_n := (\mathbb{P}^1)^n \setminus \Delta$, where Δ is the large diagonal (i.e. the n points are pairwise distinct). By normalising the first three points to $e_1 := [1 : 0]$, $e_2 := [0 : 1]$, $e_3 := [1 : 1]$, a point in $M_{0,n}$ is uniquely determined by the coordinates of the last $n - 3$ points, giving an open immersion $M_{0,n} \hookrightarrow (\mathbb{P}^1)^{n-3}$. The closure of this map is the whole of $(\mathbb{P}^1)^{n-3}$, so we could sensibly view $(\mathbb{P}^1)^{n-3}$ as a compactification of $M_{0,n}$. However, this compactification is not symmetric in the points, and hence isn't favoured as a natural compactification: not all points are treated equally when looking at degenerate configurations. Instead, we need to consider all possible morphisms $M_{0,n} \rightarrow \mathbb{P}^1$ simultaneously to get a symmetric compactification, which has a strong geometric interpretation.

Let $\tilde{\mathcal{Q}}_n$ be the set of ordered quadruples of $[n] = \{1, \dots, n\}$ ($n \geq 4$). We define the morphism $\tilde{q}_v : M_{0,n} \rightarrow \mathbb{P}^1$ for each $v \in \tilde{\mathcal{Q}}$ to be the coordinate of the fourth point after normalising the first three to e_1, e_2, e_3 . The morphism

$$\prod_{v \in \tilde{\mathcal{Q}}_n} \tilde{q}_v : M_{0,n} \hookrightarrow \prod_{v \in \tilde{\mathcal{Q}}_n} \mathbb{P}^1$$

is an open immersion (an isomorphism onto a Zariski-open subset of $(\mathbb{P}^1)^{\tilde{\mathcal{Q}}_n}$).

Definition 1. $\overline{M}_{0,n}$ is the closure of the image of the above map.

2.2 The Gerritzen-Piwek compactification $\overline{\mathbf{B}}_n$

Gerritzen and Piwek **[GP]** construct the variety $\overline{\mathbf{B}}_n$ as a direct generalisation of Definition 1. We consider the open set $U_n \subset (\mathbb{P}^2)^n$ of points in general position (configurations such that no three are colinear, i.e. don't lie in a hyperplane), and its quotient is the moduli space $\mathbf{B}_n := U_n / \mathrm{PGL}_3$. Since PGL_3 is a larger group, we need to normalise four points instead of three, to $e_1 := [1 : 0 : 0]$, $e_2 := [0 : 1 : 0]$, $e_3 := [0 : 0 : 1]$, $e_4 := [1 : 1 : 1]$, with the remaining coordinates on $(\mathbb{P}^2)^{n-4}$ uniquely representing each PGL_2 equivalence class.

Let $\mathcal{Q}_n$ be the set of ordered quintuples on $[n] := \{1, \dots, n\}$ (where $n \geq 5$). For a quintuple $v \in \mathcal{Q}_n$, we

define the map

$$q_v : \mathbf{B}_n \longrightarrow \mathbb{P}^2$$

to be the image of the point p_{v_5} after applying the projective transformation that sends the first four points $p_{v_i} \mapsto e_i$ (the coordinates of the v_5^{th} point under this representative of the equivalence class). This is a morphism, as it is the composition of a section $\mathbf{B}_n \rightarrow U_n \subset (\mathbb{P}^2)^n$ and the forgetful map projecting onto the v_5^{th} copy of $\mathbb{P}^2$.

Definition 2. (Gerritzen & Piwek, [GP]) The *Gerritzen-Piwek compactification* $\overline{\mathbf{B}}_n$ is the closure of $\mathbf{B}_n$ under the product morphism

$$\prod_{v \in \mathcal{Q}_n} q_v : \mathbf{B}_n \longrightarrow \prod_{v \in \mathcal{Q}_n} \mathbb{P}^2$$

where the product is the fibred product over $\text{Spec}(k)$. Note the map is an open immersion. In practice we can work with ordered quintuples $v \in \mathcal{Q}'_n$ where the first four elements are in increasing order, since the resulting closure is also isomorphic to $\overline{\mathbf{B}}_n$: for each permutation of the first four points, the coordinates only change by a projective transformation. We compose the maps

$$\mathbf{B}_n \xrightarrow{\prod_{v' \in \mathcal{Q}'_n} q_{v'}} \prod_{v' \in \mathcal{Q}'_n} \mathbb{P}^2 \xrightarrow{\prod_{v' \in \mathcal{Q}'_n} f_{v'}} \prod_{v \in \mathcal{Q}_n} \mathbb{P}^2$$

where $f_{v'} : \mathbb{P}^2 \rightarrow \prod_{\sigma \in S_4} \mathbb{P}^2$ is the diagonal embedding composed with the projective coordinate changes for each permutation of the first four elements of a quintuple in $\mathcal{Q}'_n$. The composition is just $\prod_{v \in \mathcal{Q}_n} q_v$, whose closure is isomorphic to the closure of the first map, because $f_{v'}$ is an isomorphism onto its image, which is the closed diagonal of $(\mathbb{P}^2)^{S_4}$. Note this process generalises directly to the case of points $\mathbb{P}^{r-1}$.

Example 3. For $\overline{\mathbf{B}}_5$, we have the open immersion $\mathbf{B}_5 \hookrightarrow (\mathbb{P}^2)^5$ since there are five quintuples in $\mathcal{Q}'_5$, corresponding to the five possible choices for the fifth point. We use coordinates $\alpha = [\alpha_1 : \alpha_2 : \alpha_3], \beta, \gamma, x, y$ on the respective copies of $\mathbb{P}^2$, corresponding to the coordinates of the p_i^{th} point ($i = 1, 2, 3, 4, 5$ respectively) after sending the other four to e_1, e_2, e_3, e_4 . The image of the equivalence class representative $[(e_1, e_2, e_3, e_4, [a : b : c])] \in \mathbf{B}_5$ is

$$\left(\left[\frac{1}{a-b} : \frac{1}{a-c} : \frac{1}{a} \right], \left[\frac{1}{b-a} : \frac{1}{b-c} : \frac{1}{b} \right], \left[\frac{1}{c-a} : \frac{1}{c-b} : \frac{1}{c} \right], \left[\frac{1}{a} : \frac{1}{b} : \frac{1}{c} \right], [a : b : c] \right).$$

Here $[a : b : c]$ ranges over all of $U := \mathbb{P}^2 \setminus V(\langle XYZ(X-Y)(X-Z)(Y-Z) \rangle) \cong \mathbf{B}_5$. The coordinates are subject to the following minimal relations, and $\overline{\mathbf{B}}_5$ is the subscheme of $(\mathbb{P}^2)^5$ cut out by them:

$$\begin{aligned} (y_1 - y_2)\alpha_1 &= (y_1 - y_3)\alpha_2 = y_1\alpha_3 \\ (y_2 - y_1)\beta_1 &= (y_2 - y_3)\beta_2 = y_2\beta_3 \\ (y_3 - y_1)\gamma_1 &= (y_3 - y_2)\gamma_2 = y_3\gamma_3 \\ y_1x_1 &= y_2x_2 &= y_3x_3. \end{aligned}$$

When $y \in \mathbb{P}^2 \setminus \{e_1, e_2, e_3, e_4\}$ there is a unique solution. And when $y = e_i$ there is a line (the line that doesn't pass through p_i or p_5) of points in the copies of $\mathbb{P}^2$ where p_j ($j \neq i, 5$) is free, i.e. a blow up of $\mathbb{P}^2$ at e_i . Hence $\overline{\mathbf{B}}_5 \cong \text{Bl}_4(\mathbb{P}^2)$, the blowup of $\mathbb{P}^2$ at four points in general linear position.

To understand Lafforgue and Kapranov's approaches and generalisations we first need to flesh out the Gelfand-MacPherson correspondence in full generality.

3 The Gelfand-MacPherson correspondence

A key result of Kapranov is that the Chow quotient $\overline{\mathbf{X}}(r, n)$ of n points in $(\mathbb{P}^{r-1})^\vee$ is isomorphic to the Chow quotient of the Grassmannian $\mathrm{Gr}(r, n)/H$ by the group $H = (k^*)^n/k^*$. The connection between configurations of n points in $(\mathbb{P}^{r-1})^\vee$ and r -dimensional linear subspaces of k^n is the Gelfand-MacPherson correspondence, and to prove it we will need to develop some theory about the Grassmannian and matroids.

It should be noted that while there is an isomorphism between point configurations in $\mathbb{P}^{r-1}$ and in $(\mathbb{P}^{r-1})^\vee$, this identification depends on a choice of isomorphism $\mathbb{P}^{r-1} \cong (\mathbb{P}^{r-1})^\vee$ (of which there is no canonical choice). As such, the modular interpretation is most correct if these cases are as distinct (but isomorphic) moduli spaces, although the issue doesn't arise for $r = 2$: points in $(\mathbb{P}^1)^\vee$ correspond to hyperplanes in $\mathbb{P}^1$, which are also points.

Definition 4. The *Grassmannian* $\mathrm{Gr}(r, n)$ is defined as the set of r -dimensional linear subspaces of k^n . It can be realised as the set of linear embeddings $F := k^r \hookrightarrow k^n$, up to a GL_r action that changes the basis vectors of F – we are parametrising linear subspaces up to equality as sets, rather than equality as linear maps. $\mathrm{Gr}(r, n)$ is another great example of a moduli space, and as a special case $\mathrm{Gr}(1, n) = \mathbb{P}^{n-1}$. The Grassmannian can be embedded as a closed subvariety of projective space through the Plücker embedding, into $\mathbb{P} := \mathbb{P}(\bigwedge^r k^n)$. The map is defined as follows

$$\mathrm{Span}\{w_1, \dots, w_r\} \mapsto [w_1 \wedge \dots \wedge w_r]$$

for any basis $\{w_1, \dots, w_r\}$ of the r -plane, with the right hand coordinate being the projective equivalence class. Not all elements of $\mathbb{P}$ can be decomposed into the wedge product of individual vectors, for example $e_1 \wedge e_2 + e_3 \wedge e_4 \notin \mathrm{Gr}(2, n)$ for any n , and the Plücker relations cut out $\mathrm{Gr}(r, n)$ as a closed subscheme of $\mathbb{P}$, although their explicit expression is not necessary for our discussion.

$\mathbb{P}$ admits a natural stratification into subschemes indexed by nonempty sets of r -element subset of $[n]$. We denote $\binom{S}{r}$ to be the set of all r -element subsets of a set S . In particular, let $U_{r,n} = \binom{[n]}{r}$ whose elements are in bijection with coordinates x_I of $\mathbb{P}$, where $x_{\{i_1, \dots, i_r\}}$ is the coefficient of $e_{i_1} \wedge \dots \wedge e_{i_r}$. For each $P \subset U_{r,n}$ we thus define two subsets

$$\begin{aligned} \mathbb{P}^{P,0} &:= \{x \in \mathbb{P} \mid x_I \neq 0 \text{ if and only if } I \in P\} \\ \mathbb{P}^P &:= \{x \in \mathbb{P} \mid x_I = 0 \text{ if } I \notin P\} = \overline{\mathbb{P}^{P,0}} \subset \mathbb{P}. \end{aligned}$$

The set of $\mathbb{P}^{P,0}$ s give a stratification of the Grassmannian into subschemes $\mathrm{Gr}^{P,0}(r, n) := \mathrm{Gr}(r, n) \cap \mathbb{P}^{P,0}$, whose closure inside $\mathrm{Gr}(r, n)$ is $\mathrm{Gr}^P(r, n) = \mathrm{Gr}(r, n) \cap \mathbb{P}^P$. To understand which strata are non-empty, we need to delve into the theory of matroids.

Definition 5. A matroid is defined by a nonempty set of bases $P \subset U_{r,n}$ satisfying the *basis exchange property*: if $A, B \in P$ and $a \in A \setminus B$, then there exists a $b \in B \setminus A$ such that $(A \cup \{b\}) \setminus \{a\} \in P$.

Matroids arise naturally in the study of configurations of n vectors indexed by $[n]$ in k^r , where $B \in P$ if and only if (the set of vectors indexed by) B spans k^r . A configuration of n general vectors (no r lie in a hyperplane) corresponds to the uniform matroid $U_{r,n}$, i.e. any r -element subset forms a basis for k^r , and other matroids correspond to degenerate configurations. An equivalent definition of a matroid includes a larger family of sets, corresponding to the collections of points that are linearly independent (of which the bases are the r -element ones), although the actual data of both conventions is equivalent.

The basis exchange property comes from the fact that if A and B index bases for k^r , then any $r - 1$

element subset $A \setminus \{a\} \subset A$ spans a hyperplane. Since B spans k^r too, at least one $b \in B$ is not contained in $A \setminus \{a\}$, and so the set $(A \setminus \{a\}) \cup \{b\}$ must also index a basis of k^r .

A configuration of n non-zero vectors uniquely determines a point configuration in $\mathbb{P}^{r-1}$, so matroids corresponding to projective point configurations satisfy an additional requirement of each $i \in [n]$ being in at least one basis, since the space spanned by $\{i\}$ can't be zero-dimensional.

For every matroid there is an associated convex polytope, which by abuse of notation we also denote P

$$P := \text{Conv} \left\{ \sum_{i \in B} e_i \mid B \text{ is a basis of } P \right\} \subset \mathbb{R}^n$$

where each e_i is a basis vector for the standard basis of $\mathbb{R}^n$. Its vertices are in bijection with the bases of P , and hence the data of a matroid is equivalent to that of its polytope. As an example, the $n-1$ -simplex $\text{Conv}\{e_i \mid i \in [n]\} \subset \mathbb{R}^n$ is the matroid polytope associated to $U_{1,n}$, and we call the class of uniform matroid polytopes $\Delta(r, n)$ the *hypersimplices* (which correspond to the matroid $U_{r,n}$).

The convex hull of a set of points is a region bounded by a set of hyperplanes, which for a matroid polytope are the hyperplanes cut out by $\sum_{i \in I} x_i = d_I^P$ for some set of integers $\{d_I^P\}_{I \subset [n]}$ (i.e. P is the set of points where $d_I^P \leq \sum_{i \in I} x_i \leq r = d_{[n]}^P$ for all I). These can be computed by

$$d_I^P = \min \left\{ \sum_{i \in I} x_i \mid (x_1, \dots, x_n) \in P \right\}.$$

Conceptually, each d_I^P is the minimum number of elements of I that must appear in a basis for P . Alternatively, for a configuration of n points in k^r , the number d_I^P is the codimension of the space spanned by the *complement* I^c . This can be seen by noting that the minimum of $\sum_{i \in I} x_i$ is equivalently the maximum of $\sum_{j \in I^c} x_j = r - \sum_{i \in I} x_i$. This maximum is achieved on integer points of P (which are vertices of P) since P is the convex hull of integer vertices, and hence any basis containing any subset of the point corresponding to I contains at most $r - d_I^P$ of them, i.e. $\text{codim}(\text{Span}(I^c)) = d_I^P$.

Any set of non-negative $\{d_I\}_{I \in [n]}$ satisfying $d_\emptyset = 0, d_{[n]} = r, d_I + d_J \leq d_{I \cup J} + d_{I \cap J}$ also equivalently defines a rank r matroid on n elements [La]. We additionally require each $d_{[n] \setminus \{i\}} \geq r - 1$ for it to correspond to a point configuration in k^r (with strict equality iff it is a projective point configuration), since the span of a single vector is either zero or one dimensional.

The following alternate definition of the matroid stratification of the Grassmannian by Lafforgue is most convenient to work with. Consider the decomposition of subspaces $k^n = E_1 \oplus \dots \oplus E_n$ where each $E_i = k$. For convenience, we let $E_I = \bigoplus_{i \in I} E_i$. In particular, note $E_\emptyset = \{0\}, E_{[n]} = k^n$.

Lemma 6. (Lafforgue, [La, I.1]) The stratum $\text{Gr}^{P,0}(r, n)$ is equivalently defined as the set of r -planes $F \subset k^n$ such that $\dim(F \cap E_I) = d_I^P$ for all $I \in [n]$.

As an example, a generic r -plane $F \subset k^n$ intersects each E_I transversally, i.e. with additive codimension (generic here means generic with respect to the graded structure of k^n). This means $\dim(F \cap E_I) = \max\{0, |I| + r - n\}$. The matroid with $d_I^P = \max\{0, |I| + r - n\}$ is the hypersimplex, and so the stratum $\text{Gr}^{\Delta(r,n),0}(r, n)$ parametrises all generic r -planes in k^n . More generally, $d_I^P - d_I^{\Delta(r,n)}$ measures how much these intersections fail to be transverse.

Lemma 7. The stratum $\text{Gr}^{P,0}(r, n)$ is empty if P is not a matroid polytope.

Proof. Given any r -plane $F \subset k^n$, we let $d_I = \dim(F \cap E_I)$. Since $E_\emptyset = \{0\}$ and $E_{[n]} = k^n$, $d_\emptyset = 0$ and $d_{[n]} = r$. And it is a fact of linear algebra that

$$\dim(F \cap E_I) + \dim(F \cap E_J) = \dim((F \cap E_I) \cup (F \cap E_J)) + \dim((F \cap E_I) \cap (F \cap E_J))$$

with the right side being $\dim(F \cap E_{I \cup J}) + \dim(F \cap E_{I \cap J})$, hence the condition $d_I + d_J \leq d_{I \cup J} + d_{I \cap J}$ is automatically satisfied, and the set of d_I s corresponds to some matroid polytope $P \subset \Delta(r, n)$, i.e. any r -plane in k^n is a point in $\text{Gr}^{P,0}(r, n)$ for some matroid polytope P . $\square$

The identification of the Grassmannian with linear embeddings $k^r \hookrightarrow k^n$ gives a collection of maps $k^r \rightarrow E_i = k$, which uniquely determine the map $k^r \hookrightarrow k^n$. Each map $k^r \rightarrow k$ can be identified with a point in $(k^r)^\vee$, giving an embedding

$$\text{Gr}(r, n) \hookrightarrow ((k^r)^\vee)^n / \text{GL}_r$$

up to the action of GL_r on k^r , which is dual to the action of GL_r on $(k^r)^\vee$. This is not surjective because not all maps are full rank, but the image of $\text{Gr}(r, n)$ is a dense open subset. To projectivise $(k^r)^\vee$, we consider a quotient of the Grassmannian.

The action of the torus $H := (k^*)^n / k^*$ on k^n (by rescaling the coordinate axes) induces an action on $\mathbb{P}$ and by extension $\text{Gr}(r, n)$. While this action is not free (consider its action on the point corresponding to $E_{[r]} \subset k^n$), it does fix each stratum and acts freely on strata $\text{Gr}^{P,0}(r, n)$ where P is a matroid corresponding to a projective point configuration. This comes from $\dim(F \cap E_{[n] \setminus \{i\}}) = r - 1$, and so the projection of any r -plane onto the i^{th} coordinate is surjective. Rescaling each axis by any set of $\alpha_i \in k^*$ gives a different subspace (also in $\text{Gr}^{P,0}(r, n)$), up to a global factor of k^* , hence the H -action on $\text{Gr}^{P,0}(r, n)$ is free when P corresponds to a projective point configuration. There is thus a smooth quotient

$$\Omega^P := \text{Gr}^{P,0}(r, n) / H \subset \mathbb{P}^{P,0} / H.$$

Definition 8. Given a matroid $P \subset \Delta(r, n)$, we define the space

$$\mathbf{X}^P(r, n) \subset ((\mathbb{P}^{r-1})^\vee)^n / \text{PGL}_r$$

to be the moduli space of n general points in $(\mathbb{P}^{r-1})^\vee$ corresponding to a matroid P – that is n -point configurations in $(\mathbb{P}^{r-1})^\vee$ such that the subsets of r points that span $(\mathbb{P}^{r-1})^\vee$ are in bijection with bases of P . In particular, the space $\mathbf{X}^{\Delta(r, n)}(r, n) := \mathbf{X}(r, n)$ is the moduli space of n general points in $(\mathbb{P}^{r-1})^\vee$. By definition (of $(\mathbb{P}^{r-1})^\vee$), this is the same as n hyperplanes in $\mathbb{P}^{r-1}$ (which is the equivalent definition some authors use).

Theorem 1. The *Gelfand-MacPherson correspondence* gives an isomorphism $\Omega^P \cong \mathbf{X}^P(r, n)$ for matroid polytopes $P \subset \Delta(r, n)$ which correspond to projective point configurations. Note that we are using the notation of [ST] and following the proof given in [La].

Proof. For any r -plane $F \in \text{Gr}^{P,0}(r, n)$, we have $\dim(F \cap E_{I^c}) = d_{I^c}^P = r - 1$ (P corresponds to a projective projection configuration). As such, the maps $F \rightarrow E_i = k$ are all surjective, and define n non-zero points $\{p_i\}_{i \in [n]} \in F^\vee$ (which are dual to the hyperplanes $\ker(F \rightarrow E_i) \subset F$).

Quotienting by the H -action on $\text{Gr}^{P,0}(r, n)$ identifies these maps up to a scalar multiple, now a point in Ω^P corresponds uniquely to n points in $\mathbb{P}(F^\vee) \cong (\mathbb{P}^{r-1})^\vee$, up to changing basis on $\mathbb{P}(F)$, i.e. up to a PGL_r action. This gives an embedding

$$\Omega^P \hookrightarrow ((\mathbb{P}^{r-1})^\vee)^n / \text{PGL}_r.$$

Now, $\dim(F \cap E_{I^c}) = \dim(\ker(F \rightarrow E_I))$ since the map $F \hookrightarrow k^n$ is an embedding. In particular, for any $I \in [n]$, the subspace U_I of $\mathbb{P}(F^\vee)$ spanned by $\{p_i\}_{i \in I}$ has codimension

$$\text{codim}(U_I) = \dim(\ker(F \rightarrow E_I)) = \dim(F \cap E_{I^c}) = d_{I^c}^P$$

by projective duality (the dimension of the intersection of several hyperplanes in $\mathbb{P}(F)$ is the same as the codimension of the subspace of $\mathbb{P}(F^\vee)$ spanned by their duals). This aligns with the expected codimensions of a projective point configuration corresponding to P .

In the other direction, any point in $\mathbf{X}^P(r, n)$ gives a collection of maps $F \rightarrow E_i$ (up to the PGL_r -action) that glue to give an r -plane $F \in \mathrm{Gr}^{P,0}(r, n)/H$, which is precisely the inverse map $\mathbf{X}^P(r, n) \rightarrow \Omega^P$. We thus have the Gelfand-MacPherson correspondence

$$\Omega^P \cong \mathbf{X}^P(r, n). \quad \square$$

In particular, Kapranov [Ka, 2.2.4] showed that for $P = \Delta(r, n)$ (the case of n points in general position), then there is an isomorphism between the Chow quotients

$$\mathrm{Gr}(r, n)/H \cong ((\mathbb{P}^{r-1})^\vee)^n // \mathrm{PGL}_r.$$

that compactify $\Omega^{\Delta(r, n)} = \mathrm{Gr}^{\Delta(r, n), 0}(r, n)/H$ and $\mathbf{X}^{\Delta(r, n)} = \mathbf{X}(r, n)$ respectively.

4 Lafforgue's compactification $\overline{\Omega}^P$

There is another way to compactify Ω^P , using the matroid stratification of the Grassmannian. This gives the so-called *Lafforgue variety*, which in the case of $P = \Delta(r, n)$ is closely related to $\overline{\mathbf{X}}(r, n)$. This compactification uses the theory of toric varieties and regular subdivisions of a matroid polytope, which give families of more degenerate r -plane configurations (and accordingly more degenerate point configurations).

Given a projective point configuration, we would like to understand what its degenerations are. Specifically, for a matroid P , we want to know which matroids Q correspond to more degenerate point configurations, where a more degenerate configuration is one where points span lower (or equal) dimensional spaces. There are two options: either $d_{[n]}^Q = d_{[n]}^P = r$, or $d_{[n]}^Q < r$. In the latter case the point configuration doesn't span $(\mathbb{P}^{r-1})^\vee$, which (while being interesting in of itself) is too degenerate to arise in the compactifications of our moduli space, since there is no free PGL_r action on a point configuration that lies entirely within a hyperplane.

As such, we are most interested in configurations with $d_{[n]}^Q = d_{[n]}^P$, and where $d_I^Q \geq d_I^P$ for all $I \in [n]$: this means that the collection of points indexed by I^c has at least as high codimension. In particular, if $Q \neq P$ then one of these inequalities is strict, and so the configuration is strictly more degenerate. When $d_I^Q \geq d_I^P$, as polytopes $Q \subset P$, and the corresponding point configuration looks like that of P , but the sets of points indexed by vertices in $P \setminus Q$ are collapsed onto hyperplanes (or lower dimension), rather than spanning $\mathbb{P}^{r-1}$, with all such degenerations obtained by matroid subpolytopes $Q \subset P$.

Instead of simply collapsing points onto a hyperplane, we'd like to retain some information about how these degenerations arise as limiting cases of configurations in $\mathrm{Gr}^{P,0}(r, n)$. Under a specific PGL_r equivalence class, any set of $r+1$ general points can be fixed to the basis vectors e_i (and $e_{r+1} := [1 : \cdots : 1]$), in particular any r linearly independent points can be fixed to the basis vectors e_i , and so no limiting sequence of configurations could ever collapse all n points to a hyperplane (concretely justifying why configurations with $d_{[n]}^Q < r$ shouldn't arise in our compactifications).

Since any subpolytope $Q \subset P$ has at least one less vertex than P , at least one basis set is collapsed to a hyperplane, which shouldn't arise under a limit through an appropriate choice of equivalence class, since they can be permanently separated with a choice of coordinates. To resolve this, instead of considering subpolytopes individually, we look at matroid subdivisions of P , and consider all the respective point configurations together, which gives a resulting divisor akin to a blow up.

4.1 Faces and subdivisions of matroid polytopes

Definition 9. A *matroid subdivision* of a matroid polytope P is a set $\underline{P}$ of matroid subpolytopes (cells) $C \subset P$ such that (a) the intersection of any two cells is a face of both, (b) the union of all cells in $\underline{P}$ is equal to P . Note a face of a matroid polytope P is given by the intersection of P with finitely many hyperplanes $\sum_{i \in I} x_i = d_I^P$ and (c) all the faces of a cell in $\underline{P}$ are also in $\underline{P}$.

The specific class of subdivisions that correspond to degenerations are *regular* subdivisions. Each regular subdivision $\underline{P}$ arises from a real-valued function w on the vertex set of P . The convex hull of the rays

$$\left\{ \sum_{i \in I} e_i \right\} \times \left\{ x \in \mathbb{R} \mid x \geq w(I) \right\} \subset \mathbb{R}^n \times \mathbb{R}$$

ranging over all vertices I of P defines a region of $\mathbb{R}^{n+1}$, whose finite faces define the cells of $\underline{P}$ upon projection to $\mathbb{R}^n$. Besides being useful in Lafforgue's compactification, regular subdivisions of $\Delta(r, n)$ also index the stratification of Kapranov's closely related Chow quotient $\overline{\mathbf{X}}(r, n)$ [Ka, 1.2].

To understand the meaning of matroid subdivisions in point configurations, we first need to understand the possible maps between strata of the Grassmannian. There is always a forgetful morphism $\mathbb{P}^{P,0} \rightarrow \mathbb{P}^{Q,0}$ for all $Q \subset P$, however this doesn't necessarily restrict to a morphism $\text{Gr}^{P,0}(r, n) \rightarrow \text{Gr}^{Q,0}(r, n)$, except in the following case:

Lemma 10. When Q is a face of P the face morphism $\text{Gr}^{P,0}(r, n) \rightarrow \mathbb{P}^{Q,0}$ is a morphism onto $\text{Gr}^{Q,0}(r, n)$.

Proof. We first consider facets (faces of codimension one), which come from the intersecting P with a single hyperplane $\sum_{i \in I} x_i = d_I^P$. Consider the map $\text{Gr}(r, n) \rightarrow \text{Gr}^{Q,0}(r, n)$ defined by

$$F \mapsto (F \cap E_I) \oplus F / (F \cap E_I) \subset E_I \oplus k^n / E_I \cong k^n$$

where the last isomorphism is the canonical identification of $k^n / E_I = (E_I \oplus E_{I^c}) / E_I$ with E_{I^c} . In the Plücker embedding, F is represented by the point $\mathbb{P}(\bigwedge^r F) \subset \mathbb{P}(\bigwedge^r k^n)$. This is indeed a point as F is r -dimensional, so there is a single non-zero r -fold wedge product of vectors in F (up to a factor of k , which is removed by the projective equivalence class). Its image is the point

$$\mathbb{P}\left(\bigwedge^{d_I^P} (F \cap E_I) \otimes \bigwedge^{r-d_I^P} (F / (F \cap E_I))\right) \subset \mathbb{P}\left(\bigwedge^r (E_I \oplus E_{I^c})\right)$$

which again is a point as it is the projective equivalence class of the tensor product of two lines, which is one-dimensional (and zero-dimensional as a projective equivalence class).

We can similarly define the above map for each basis vector $\bigwedge^r E_J$ of $\mathbb{P}^P$ (i.e. J is a vertex of P). It gets sent to $\bigwedge^{d_I^P} E_{I \cap J} \otimes \bigwedge^{r-d_I^P} E_{J \setminus I}$ (up to projective equivalence). The exterior algebra $\bigwedge^m V$ of any vector space V is zero if $m > \dim V$, and either the left or right component of the tensor product has too large dimension if $\dim(E_{I \cap J}) = |I \cap J| \neq d_I^P$, meaning $\bigwedge^r E_J \mapsto 0$. Now, since F is cut out by the hyperplane $\sum_{i \in I} x_i = d_I^P$, then $|I \cap J| = d_I^P$ if and only if $J \in F$, so $\bigwedge^r E_J \mapsto 0$ if $J \notin Q$, and if $J \in Q$ then $\bigwedge^r E_J \mapsto \bigwedge^r E_J$. Extending these maps by k -linearity gives a map between projective exterior algebras $\mathbb{P}^P \rightarrow \mathbb{P}^Q$, which is exactly the forgetful morphism.

The image of $\text{Gr}^{P,0}(r, n)$ is contained in $\text{Gr}^{Q,0}(r, n)$, and so (for codimension one) the face map is indeed a morphism $\text{Gr}^{P,0}(r, n) \rightarrow \text{Gr}^{Q,0}(r, n)$. Faces of higher codimension can be mapped to by repeatedly composing these forgetful morphisms, and so the face maps $\text{Gr}^{P,0}(r, n) \rightarrow \text{Gr}^{Q,0}(r, n)$ are both well-defined and morphisms for any face Q of P . $\square$

It is easy to verify that the face map is H -equivariant (as the forgetful morphism $\mathbb{P}^{P,0} \rightarrow \mathbb{P}^{Q,0}$ is), and so gives a morphism between point configurations

$$\mathbf{X}^P(r, n) \longrightarrow \mathbf{X}^Q(r, n).$$

Its modular interpretation is as follows: cutting out a codimension one face $Q \subset P$ by the hyperplane $\sum_{i \in I} x_i = d_I^P$ imposes the restriction that exactly d_I^P many elements of I appear in any basis of Q , i.e. the set of I points. If $d_I^P = 0$ then this is a restriction of the forgetful morphism $((\mathbb{P}^{r-1})^\vee)^n / \mathrm{PGL}_r \rightarrow ((\mathbb{P}^{r-1})^\vee)^{n-|I|} / \mathrm{PGL}_r$ forgetting the coordinates of the points corresponding to I .

If $d_I^P = 1$, the morphism identifies the points corresponding to I in $(\mathbb{P}^{r-1})^\vee$ to a single point (i.e. identifies the points in I), and more generally the points corresponding to I are projected to lie on a subspace of dimension $d_I^P - 1$ inside $(\mathbb{P}^{r-1})^\vee$. Face maps of higher codimension behave identically, projecting multiple sets of points onto multiple respective (possibly intersecting) subspaces of $(\mathbb{P}^{r-1})^\vee$. This gives us the necessary tools to study the configuration spaces corresponding to subdivisions:

Definition 11. For a matroid subdivision $\underline{P}$ of P , we can define a subscheme of $\mathbb{P}^{P,0}$ to be the set of points $x \in \mathbb{P}^{P,0}$ such that the projection $\pi_{P \rightarrow C} : \mathbb{P}^{P,0} \rightarrow \mathbb{P}^{C,0}$ maps into $\mathrm{Gr}^{C,0}(r, n)$ for each $C \in \underline{P}$. Explicitly, it is the subscheme

$$\mathrm{Gr}_{\underline{P}}^{P,0}(r, n) := \bigcap_{C \in \underline{P}} \pi_{P \rightarrow C}^{-1} \left(\mathrm{Gr}^{C,0}(r, n) \right) \subset \mathbb{P}^{P,0}.$$

Note the existence of the face maps means $\mathrm{Gr}_{\underline{P}}^{P,0}(r, n)$ is uniquely determined by the maximal cells of $\underline{P}$.

If $\underline{P} = \{P\}$ is the trivial subdivision then $\mathrm{Gr}_{\underline{P}}^P(r, n) = \mathrm{Gr}^{P,0}(r, n)$, and more generally $\mathrm{Gr}_{\underline{P}}^{P,0}(r, n)$ has a modular interpretation, classifying families of r -planes F_C over $\mathrm{Gr}^{C,0}(r, n)$ for all cells $C \in \underline{P}$. These families are equipped with a natural correspondence from the face maps: over a point $x \in \mathrm{Gr}_{\underline{P}}^{P,0}(r, n)$, the plane in the family $F_{C \cap C'}$ is the image of the planes in $F_C, F_{C'}$ under the respective face morphisms (for any $C, C' \in \underline{P}$ where $C \cap C'$ is nonempty).

The Gelfand-MacPherson correspondence gives a natural identification between families of r -planes F_C over $\mathrm{Gr}^{C,0}(r, n)$ and families $\tilde{F}_C$ of projective point configurations, after quotienting by H . While H acts freely on each family F_C , there isn't an action of $\bigoplus_{C \in \underline{P}} H$ on $\mathrm{Gr}_{\underline{P}}^{P,0}(r, n)$ since the actions on families must be compatible with the face maps between them. We denote the group of compatible actions by $H_{\underline{P}}$, of which a more precise definition is given in Lemma 13. This group acts as H on each F_C where C is a maximal cell, and then the $H_{\underline{P}}$ action on $F_{C'}$ for each face $C' \subset C$ is entirely determined by the $H_{\underline{P}}$ action on F_C and the face map $\mathrm{Gr}^{C,0}(r, n) \rightarrow \mathrm{Gr}^{C',0}(r, n)$.

We claim that given two families of point configurations $\tilde{F}_C, \tilde{F}_{C'}$, the family $\tilde{F}_{C \cap C'}$ should then carry the data of the ways configurations in $\tilde{F}_C$ and $\tilde{F}_{C'}$ can be compatibly glued as a degeneration of $\mathbf{X}^P(r, n)$. The quotient $\mathrm{Gr}_{\underline{P}}^{P,0}(r, n)/H_{\underline{P}}$ then has an interpretation as a configuration space of point configurations on a gluing of several copies of $\mathbb{P}^{r-1}$ that arises from a degeneration of $\mathbf{X}^P(r, n)$.

4.2 The Lafforgue variety

We can now introduce the projective scheme $\overline{\Omega}^P$, a compactification of Ω^P . There is a natural embedding

$$\Omega^P = \mathrm{Gr}^{P,0}(r, n)/H \subset \mathcal{A}_0^P := \mathbb{P}^{P,0}/H$$

into the torus $\mathcal{A}_0^P$, and the compactification we seek arises as a projective subscheme of a toric variety $\mathcal{A}^P$ compactifying this torus.

Definition 12. Given a matroid polytope P we can construct the normal fan to its secondary polytope, whose cones correspond to subdivisions of P . The fan of cones that correspond to regular subdivisions by matroid polytopes gives the *Lafforgue toric variety* $\mathcal{A}^P$, with maximal torus $\mathcal{A}_0^P$.

As an example, for $P = \Delta(2, 4)$ the polytope is the octahedron. It admits three partial subdivisions and three complete triangulations (all regular), with its secondary polytope isomorphic to the triangle. The

normal fan is that of $\mathbb{P}^2$, i.e. $\mathcal{A}^{\Delta(2,4)} \cong \mathbb{P}^2$: each partial subdivision gives a copy of $\mathbb{P}^1$, glued to the other $\mathbb{P}^1$ s at two points (corresponding to their common triangulations).

More simply, $\mathcal{A}^P$ is a toric variety stratified by subtori $\mathcal{A}_{\underline{P}}^P$ for each regular matroid subdivision $\underline{P}$ of P , with $\mathcal{A}_{\underline{P}}^P \subset \overline{\mathcal{A}_{\underline{P}'}}^P$ if and only if $\underline{P}$ is a refinement of $\underline{P}'$.

Theorem 2. (Lafforgue, [La, II.4]) There exists a projective subscheme $\overline{\Omega}^P \subset \mathcal{A}^P$ equipped with a quotient morphism

$$\overline{\Omega}^P \longrightarrow \mathcal{A}^P / \mathcal{A}_0^P$$

that is the restriction of the quotient morphism $\mathcal{A}^P \longrightarrow \mathcal{A}^P / \mathcal{A}_0^P$. This gives a natural stratification of $\overline{\Omega}^P$, into fibres $\overline{\Omega}_{\underline{P}}^P$ over each point $\mathcal{A}_{\underline{P}}^P / \mathcal{A}_0^P$ (the orbits of the quotient). These fibres have a particularly nice description:

- The open stratum $\overline{\Omega}_{\{P\}}^P$ is isomorphic to the quotient $\text{Gr}^{P,0}(r,n)/H = \Omega^P$, i.e. $\overline{\Omega}^P$ contains Ω^P as an open subset.
- The other strata $\overline{\Omega}_{\underline{P}}^P \cong \text{Gr}_{\underline{P}}^{P,0}(r,n)/H_{\underline{P}}$, where $H_{\underline{P}}$ is the subtorus of $\mathbb{P}^{P,0}$ that stabilises $\mathcal{A}_{\underline{P}}^P$. In particular $H_{\{P\}} = H$, and so the open stratum being Ω^P is a special case of this property.

The existence of this subscheme is nontrivial, and so its proof won't be repeated here, but the motivating reason for why it should exist (and why we construct it) comes from the interpretation of $\text{Gr}_{\underline{P}}^{P,0}(r,n)/H_{\underline{P}}$ as a space of degenerations.

Importantly, there is an isomorphism $\overline{\Omega}^{\Delta(2,n)} \cong \overline{M}_{0,n}$. Generally, $\overline{\Omega}^P$ is not always a variety as it may be reducible, in particular Keel and Tevelev show that $\overline{\Omega}^{\Delta(3,n)}$ is reducible for sufficiently large n , which has the consequence $\overline{\Omega}^{\Delta(3,n)} \not\cong \overline{X}(3,n)$, $\overline{B}_n$ [KT, 3.10]. Since $\overline{\Omega}^P$ isn't always irreducible, it may be a stretch to call it a compactification, and instead the closure of the open stratum Ω^P (which we call the main stratum) inside $\overline{\Omega}^P$ could be considered the appropriate irreducible compactification. We denote the main stratum of $\overline{\Omega}^{\Delta(r,n)}$ by $\overline{X}_L(r,n)$.

Lemma 13. The face map $\Omega^P \rightarrow \Omega^Q$ extends to a morphism $\overline{\Omega}^P \rightarrow \overline{\Omega}^Q$ for all faces $Q \subset P$.

Proof sketch. Given a face $Q \subset P$, and any regular subdivision $\underline{P}$ of P , there is a regular subdivision $\underline{Q}$ given by the intersections $C \cap Q$ for all cells $C \in \underline{P}$. The following diagram of face maps commutes

$$\begin{array}{ccc} \mathbb{P}^{P,0} & \longrightarrow & \mathbb{P}^{Q,0} \\ \downarrow & & \downarrow \\ \text{Gr}^{C,0}(r,n) & \longrightarrow & \text{Gr}^{C \cap Q,0}(r,n) \end{array}$$

for all cells $C \in \underline{P}$, giving a well-defined forgetful face map on the restriction to $\text{Gr}_{\underline{P}}^{P,0}(r,n) \rightarrow \text{Gr}_{\underline{Q}}^{Q,0}(r,n)$. This map is $H_{\underline{P}}$ equivariant, where $H_{\underline{P}}$ acts on $\text{Gr}_{\underline{Q}}^{Q,0}(r,n)$ via its quotient $H_{\underline{Q}}$, and hence the face map gives a morphism between strata

$$\text{Gr}_{\underline{P}}^{P,0}(r,n)/H_{\underline{P}} \rightarrow \text{Gr}_{\underline{Q}}^{Q,0}(r,n)/H_{\underline{Q}}.$$

5 Morphisms between compactifications of configurations in $\mathbb{P}^2$

The varieties $\overline{B}_n$, $\overline{X}(3,n)$ and $\overline{X}_L(3,n)$ are more closely related than merely being birational on the open subset of general point configurations. As a special case, we have

Example 14. There is an isomorphism $\overline{\mathbf{B}}_5 \cong \overline{\mathbf{X}}(3, n) \cong \overline{\Omega}^{\Delta(3, n)} \cong \overline{\mathbf{M}}_{0,5}$.

Proof. The compactification $\overline{\mathbf{M}}_{0,5}$ is obtained by adjoining ten irreducible divisors D_I , each isomorphic to $\mathbb{P}^1$ indexed by $I \in \binom{[5]}{2}$. For $I \neq J$, D_I and D_J intersect if and only if $I \cap J = \emptyset$, with this intersection at a point. Kapranov describes a collection of birational morphisms $\sigma_i : \overline{\mathbf{M}}_{0,5} \rightarrow \mathbb{P}^2$ for $1 \leq i \leq 5$, which is an isomorphism away from the divisors $\{D_I\}_{i \in I}$, and contracts these four divisors to a point [Ka, 4.2].

Geometrically, σ_i forgets the adjoined limits corresponding to p_i, p_j approaching ($i \neq j$), and retains all other information. Since each point $x \in \overline{\mathbf{M}}_{0,5}$ is contained in at most two divisors, for at least one i the map σ_i is an isomorphism from an open set containing x onto its image. As such, the product morphism

$$\prod_{i=1}^5 \sigma_i : \overline{\mathbf{M}}_{0,5} \hookrightarrow \prod_{i=1}^5 \mathbb{P}^2$$

is a closed embedding. On $\mathbf{M}_{0,5} \cong \mathbf{B}_5$, the i^{th} Kapranov map σ_i is the same as the map $q_{v'} : \mathbf{B}_5 \rightarrow \mathbb{P}^2$, where $v' \in \mathcal{Q}'_5$ is the quintuple with fifth entry equal to i . This gives an embedding

$$\prod_{v' \in \mathcal{Q}'_5} q_{v'} : \mathbf{B}_5 \hookrightarrow \overline{\mathbf{M}}_{0,5} \subset (\mathbb{P}^2)^5$$

with $\overline{\mathbf{B}}_5$ being the closure of this map. Since $\overline{\mathbf{M}}_{0,5}$ is closed and the image of $\mathbf{B}_5$ is dense in $\overline{\mathbf{M}}_{0,5}$, then $\overline{\mathbf{B}}_5 \cong \overline{\mathbf{M}}_{0,5}$. By projective duality of Grassmannian, $\overline{\mathbf{X}}(2, 5) \cong \overline{\mathbf{X}}(3, 5)$ and $\overline{\Omega}^{\Delta(2,5)} \cong \overline{\Omega}^{\Delta(3,5)}$, the former two spaces being isomorphic to $\overline{\mathbf{M}}_{0,5}$, hence as desired

$$\overline{\mathbf{B}}_5 \cong \overline{\mathbf{X}}(3, n) \cong \overline{\Omega}^{\Delta(3, n)} \cong \overline{\mathbf{M}}_{0,5}. \quad \square$$

There is a more general relationship, to see this we first need to understand what it means for a variety to be normal.

Definition 15. An variety X over a field is said to be *normal* if any finite birational morphism $Y \rightarrow X$ is an isomorphism. Here finite means the morphism has finite fibres, i.e. the preimage of any point is a finite set. Not all birational morphisms are finite: as a simple example the blow up of the projective plane at a point $\text{Bl}_{\{p\}}(\mathbb{P}^2) \rightarrow \mathbb{P}^2$ has a fibre isomorphic to $\mathbb{P}^1$ over $\{p\}$.

To any variety X , we can associate its *normalisation*, which is a variety X^ν equipped with a finite birational morphism $X^\nu \rightarrow X$. It has the nice property that all finite birational morphisms $Y \rightarrow X$ factor uniquely through X^ν . As such, two varieties having the same normalisation is a fairly strong condition, and is much stronger than merely being birational.

Theorem 3. (Schaffler & Tevelev, [ST, 1.1]) There is a finite birational morphism $\overline{\mathbf{X}}(3, n) \rightarrow \overline{\mathbf{B}}_n$.

Proof. The forgetful morphism $\overline{\mathbf{X}}(3, n) \rightarrow \prod_{I \in \binom{[n]}{5}} \overline{\mathbf{X}}(3, I)$ is finite and birational [ST, 3.5]. In particular, since $\overline{\mathbf{X}}(3, I) \cong \overline{\mathbf{X}}(2, I) = \overline{\mathbf{M}}_{0,I}$ then we get the following commutative diagram:

$$\begin{array}{ccccccc} \mathbf{X}(3, n) & \hookrightarrow & \overline{\mathbf{X}}(3, n) & \longrightarrow & \prod_{I \in \binom{[n]}{5}} \overline{\mathbf{X}}(3, I) & \xrightarrow{\cong} & \prod_{I \in \binom{[n]}{5}} \overline{\mathbf{M}}_{0,I} \\ \downarrow \cong & & & & & & \downarrow \\ \mathbf{B}_n & \hookrightarrow & \overline{\mathbf{B}}_n & \hookrightarrow & \prod_{\mathcal{Q}'_n} \mathbb{P}^2. & & \end{array}$$

For convenience, denote the composition of the top maps by f . $\mathbf{X}(3, n)$ is a dense open subset of $\overline{\mathbf{X}}(3, n)$, and the composition of the finite birational morphism and the closed embedding is a finite birational morphism, and a closed map, which restricts to an isomorphism on $\mathbf{X}(3, n)$.

Because it is a closed map, the image of $f(\overline{\mathbf{X}}(3, n))$ is contained in the closure $\overline{f(\mathbf{X}(3, n))}$. And by definition, $\overline{\mathbf{B}}_n = \overline{f(\mathbf{X}(3, n))}$, hence $\overline{\mathbf{X}}(3, n)$ maps finitely and birationally into $\overline{\mathbf{B}}_n = \overline{f(\mathbf{X}(3, n))}$. $\square$

As a corollary, Keel and Tevelev show that there exists a finite birational morphism $\overline{\mathbf{X}}_L(3, n) \rightarrow \overline{\mathbf{X}}(3, n)$ [KT, 2.9], and so the corresponding normalisations are isomorphic:

$$\overline{\mathbf{X}}_L(3, n)^\nu \cong \overline{\mathbf{X}}(3, n)^\nu \cong \overline{\mathbf{B}}_n^\nu.$$

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